

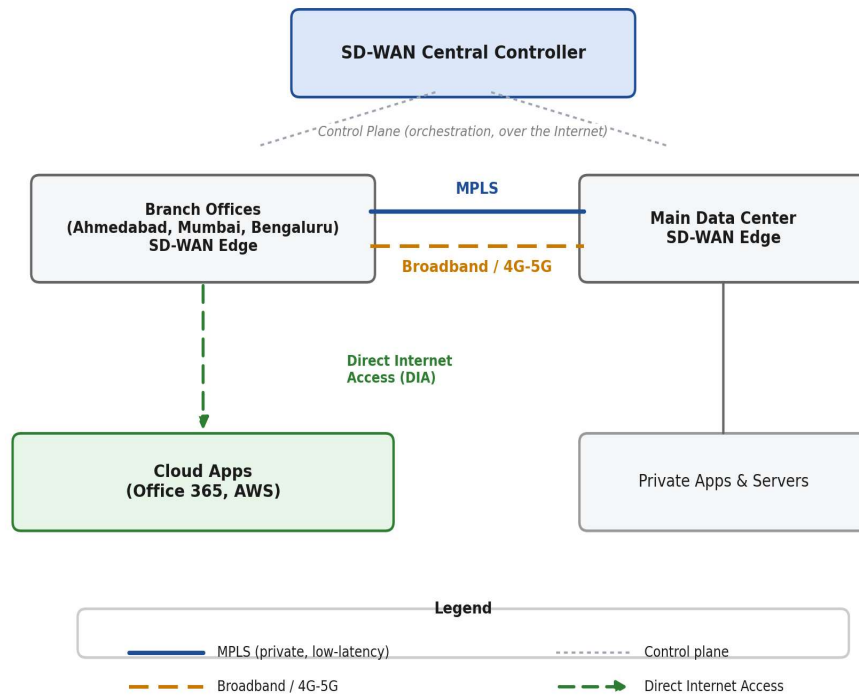
Hybrid WAN (SD-WAN)

Task 1: Create a comparison table listing at least 5 key differences between traditional WAN and SD-WAN, focusing on how data is routed, managed, and secured.

Feature / Aspect	Traditional WAN	SD-WAN (Software-Defined Wide Area Network)
Control Architecture	Decentralized: Each router control plane operates independently; configuration requires individual CLI commands per device.	Centralized: A centralized SD-WAN controller manages policies and configurations dynamically across all endpoints.
Data Routing & Traffic Management	Static/Packet-Based: Routes traffic using fixed rules, standard routing protocols (e.g., BGP, OSPF), or static IP addresses.	Application-Aware & Dynamic: Continuously measures network performance (latency, jitter, packet loss) and dynamically routes traffic on the optimal link.
Transport Mediums	Relies heavily on costly, dedicated private links like MPLS leased lines.	Transport-Agnostic: Combines and load-balances multiple link types simultaneously (MPLS, Commercial Broadband, 4G/5G, LTE).
Security Architecture	Backhauls all branch traffic through a central data center for inspection before reaching the Internet (creates bottlenecks).	Integrated Direct Internet Access (DIA) with end-to-end encryption (IPsec) and integrated cloud firewall/SASE security.
Provisioning & Management	Manual & Slow: Requires manual on-site setup, long carrier lead times, and complex command-line configurations.	Zero-Touch Provisioning (ZTP): Edge devices automatically download configurations from the central controller upon connecting to the internet.

Task 2: Draw a simple network diagram (on paper or using a tool like draw.io) showing how SD-WAN connects multiple branch offices to cloud apps (like Office 365 or AWS) and the main data center. Label the SD-WAN controllers, edges, and at least two types of WAN links (e.g., MPLS, broadband).

SD-WAN Network Architecture (Simplified)



Key Diagram Components to Label:

- **SD-WAN Central Controller:** Orchestrates policies across all connected devices and provides a single pane of glass for management.
- **SD-WAN Edge Devices:** Placed at each branch location and the Main Data Center to apply routing and security policies.
- **WAN Links (Underlay):** Connects branches using a hybrid mix of MPLS - Multiprotocol Label Switching (for critical, low-latency traffic) and Broadband/5G (for general internet/cloud access).
- **Cloud Destinations:** Direct Internet Access (DIA) routes cloud traffic (Office 365, AWS) directly from branch SD-WAN edge devices without backhauling.

Task 3: *Pick any app you use daily (like WhatsApp, Zomato, or Instagram) and write a short scenario explaining how SD-WAN could improve user experience for that app when used across multiple cities in India.*

Hint: *Think about speed, reliability, and security when users access the app from different locations.*

App chosen: Zomato (food delivery)

The situation: Zomato has offices in cities like Delhi, Mumbai, Bengaluru, and Ahmedabad. These offices constantly send and receive data — order details, delivery locations, and payment information — to and from Zomato's main servers.

How SD-WAN helps, in simple terms:

- **Faster updates:** SD-WAN picks the quickest available internet connection for Zomato's offices to send data to its servers. This means order and delivery information reaches the servers faster, so the map and order status in the app update smoothly for the customer.
- **Fewer outages:** If one internet connection at an office stops working or slows down, SD-WAN automatically switches to a backup connection right away. So even during busy times, like lunch hour or a festival sale, Zomato's systems keep running without customers noticing any interruption.
- **Better security:** SD-WAN encrypts the data traveling between Zomato's offices and its servers, which helps keep information like payment details safe.

Clarification: SD-WAN only manages the network inside Zomato's own offices and servers — it doesn't reach a customer's phone. The customer's phone connects to Zomato's servers over the regular internet/mobile network. So SD-WAN speeds things up "behind the scenes," which is what makes the app feel fast and reliable to the person using it.

Task 4: *List and explain three key benefits of SD-WAN for a company with offices in Ahmedabad, Mumbai, and Bengaluru. For each benefit, give a real-world example of how it would help employees or customers.*

1. Dynamic Traffic Path Selection & Performance Optimization

Explanation: SD-WAN monitors link quality in real time and automatically steers traffic over the best path based on network conditions and application priority.

Real-World Example: Employees in the Mumbai office attending an HD video conference with team members in Bengaluru will have their voice/video packets routed over a high-priority, jitter-free MPLS line. Simultaneously, non-critical web browsing or email sync will run over a standard broadband connection, preventing video freeze or call drops.

2. Cost Reduction via Transport Independence (Hybrid Networking)

Explanation: Companies can reduce reliance on expensive private MPLS lines by pairing cost-effective commercial fiber broadband or 5G connections without compromising reliability or performance.

Real-World Example: The branch office in Ahmedabad can replace secondary expensive backup lines with dual high-speed local broadband connections managed by an SD-WAN edge router, cutting recurring network bandwidth costs while maintaining zero downtime for employees accessing company systems.

3. Simplified Management and Fast Branch Deployment (Zero-Touch Provisioning)

Explanation: IT managers can deploy, configure, and update network policies across all branches remotely from a central console without needing dedicated on-site network engineers.

Real-World Example: When opening a new branch office or outlet in Bengaluru, the company ships an SD-WAN edge device to the site. A local staff member plugs in power and internet cables, and the device automatically downloads its full network configuration and security policies from the central management controller within minutes, allowing immediate operations.